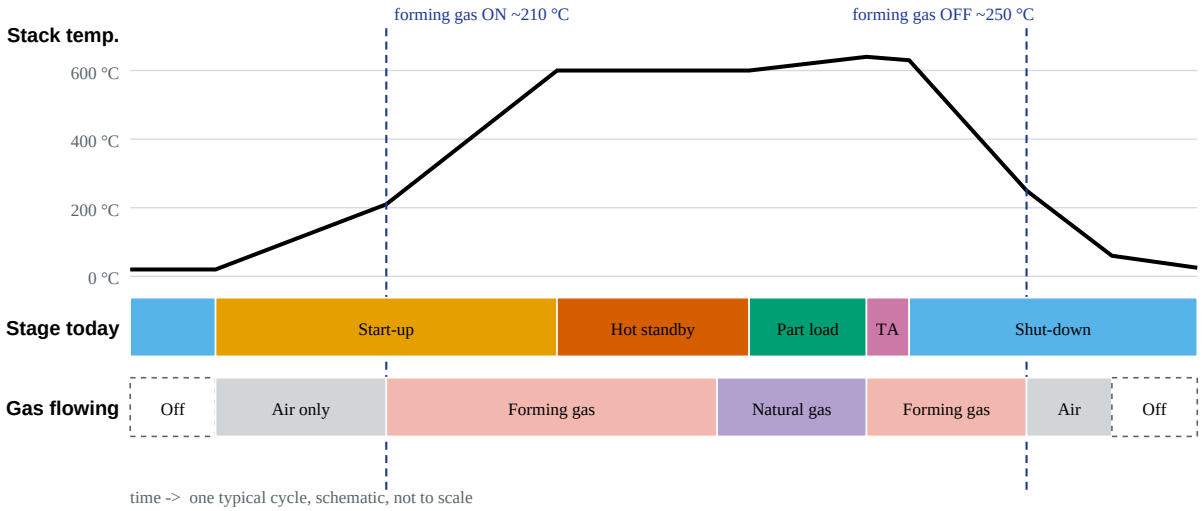
Can the gas flows redefine the plant's stages?

Yes, partly. The gas valves switch at clear, repeatable moments, so they can add sharp borders that today's temperature rules don't use. Gas alone can't replace the temperature rules, though: some different stages run on exactly the same gas.

Today a stage is decided mainly from the stack temperature, plus heater power and electric current. The gas valves are what the operator actually switches by hand. Borders taken from them are sharper and easy to check against the operator's log.

Contents: 1 The answer · 2 What the gas shows · 3 Two ways to redefine · 4 Current and power · 5 After you choose

1 · One start-to-stop cycle



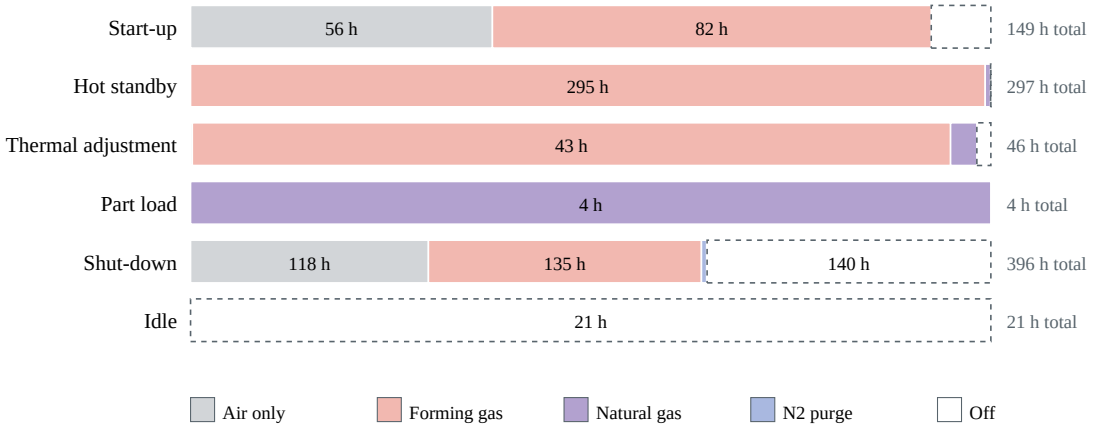
The dashed lines mark where the gas changes but today's stage does not. Forming gas comes on at about 210 °C while heating up, and goes off at about 250 °C while cooling down. Those are the two new borders on offer.

Air only: fans blow air through the cathode (the air side of the cells). **Forming gas:** nitrogen with a little hydrogen (1,200 + 60 litres per minute); it protects the fuel side of the cells from rusting while hot. **Natural gas:** the real fuel, only just before and during power. **Off:** nothing flows.

If you remember one thing: the gas gives two sharp new borders, one on the way up and one on the way down. Temperature still decides which stage you are in.

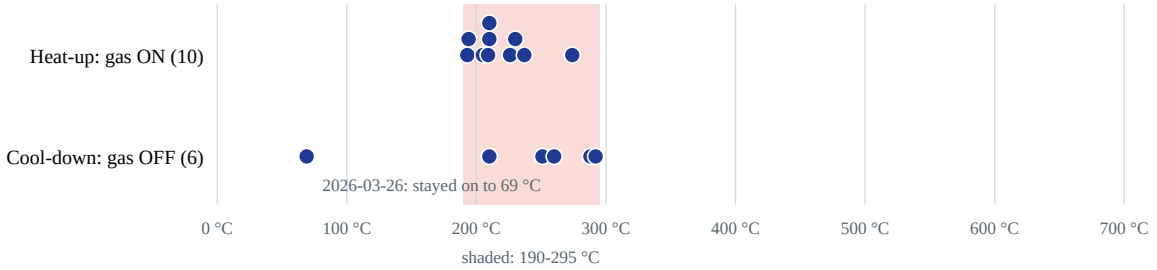
2 · What the gas lanes show

Each stage sits on a typical gas. Start-up and shut-down are the exceptions: each is split almost evenly between **air** and **forming gas**, and the switch happens at a steady temperature.



All 39 day files, minute by minute. Bar length is the share of that stage's hours; the label gives the hours. Gas from the flow controllers (air FIRC 111+112, N2 FIRC 13, H2 FIRC 15, natural gas FIRC 12).

Hot standby = forming gas: 295 of 297 hours; the other 2 hours are the natural-gas switch-over before power. **Part load = natural gas:** all 4.5 hours. **Start-up:** 56 h in air, 82 h under forming gas. **Shut-down:** 118 h in air, 135 h under forming gas, 140 h with nothing flowing while it cools on its own.



Going up, forming gas came on between 193 and 274 °C in all 10 heat-ups. Going down, it went off between 210 and 292 °C in 5 of 6 cool-downs. Once (2026-03-25 to 26) it was left on almost to room temperature.

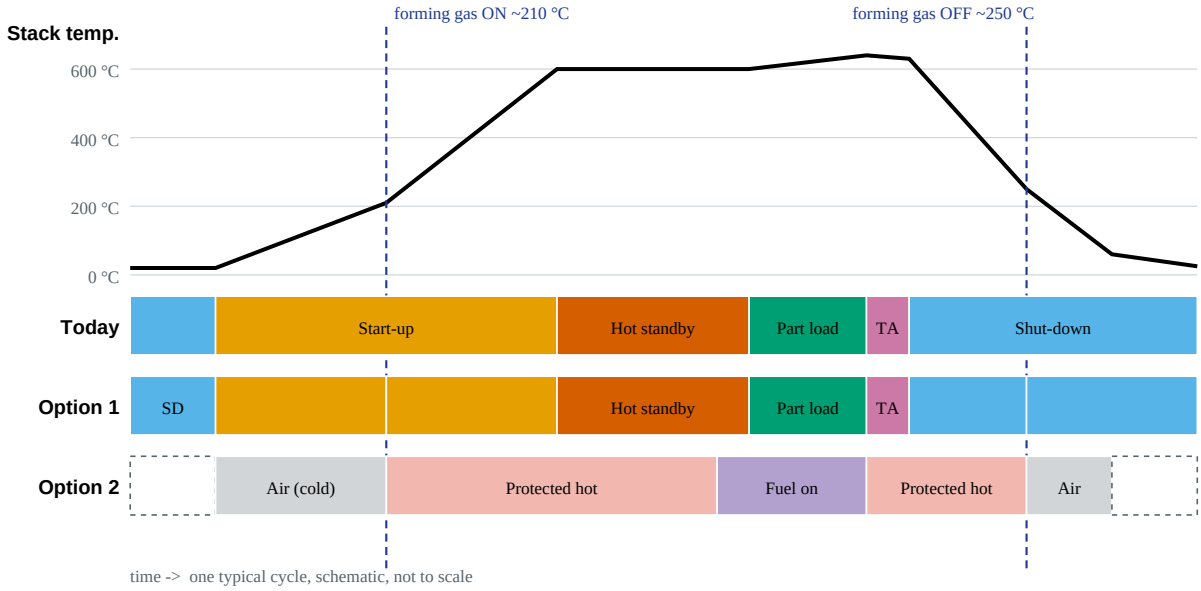
What gas can't tell apart

Hot standby vs thermal adjustment: both run on forming gas; only the temperature shows holding versus moving. **Idle vs end of shut-down vs plant off:** nothing flows in all three; only the history tells them apart. Overlaps of natural gas, nitrogen and hydrogen last 10-140 seconds during the switch to natural gas: too short to be a stage.

If you remember one thing: forming gas comes on at about 200 °C going up and goes off at about 250 °C coming down. That held every time except once.

3 · Two ways to redefine the stages

Option 1 keeps today's stages and adds the two gas borders inside them. **Option 2** names the stages after the gas and uses temperature only for detail. Recommended: option 1.



Option 1 changes only the start-up and shut-down rows: each is cut in two at the gas switch. Option 2 swaps the whole row for four gas names: Plant off, Air, Protected hot, Fuel on.

Option 1 (recommended): same six stages. Start-up gets *heat-up in air* and *heat-up under forming gas*; shut-down gets *cool-down under forming gas* and *cool-down in air*. The stage counts, the synthetic data and the 50-sensor stage checker keep working.

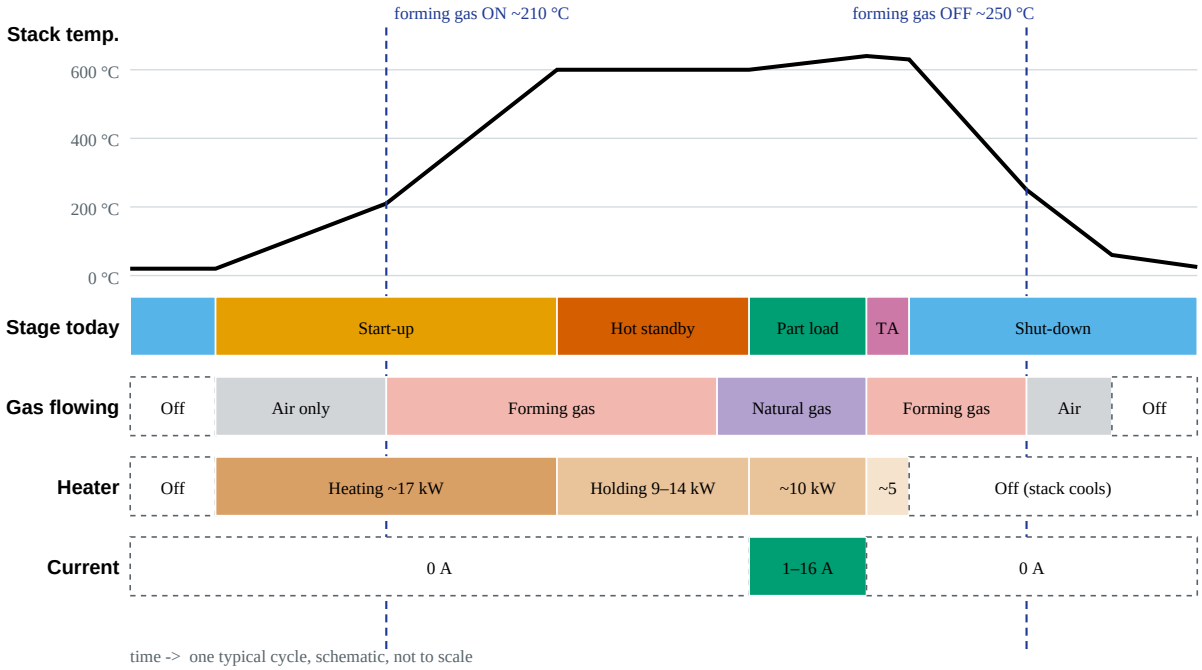
Option 2: four main stages from the gas: *Plant off*, *Air*, *Protected hot*, *Fuel on*. Heating, holding or cooling becomes the second level. Simple to explain to operators; hot standby and thermal adjustment merge into one stage.

	Option 1	Option 2
Main stage list	Unchanged (6 stages)	New (4 gas stages)
Rules to change	One new border rule	Most rules rewritten
Stage counts, notebook 9.1-9.3	Unchanged	Recomputed, not comparable
Synthetic data, 50-sensor checker (86 %)	Keep working	Rebuild and retest
Decision of 2026-10-15 (six stages stay)	Kept	Reversed

If you remember one thing: option 1 adds two borders and breaks nothing. Option 2 is cleaner on paper but means redoing everything that is built on the stages.

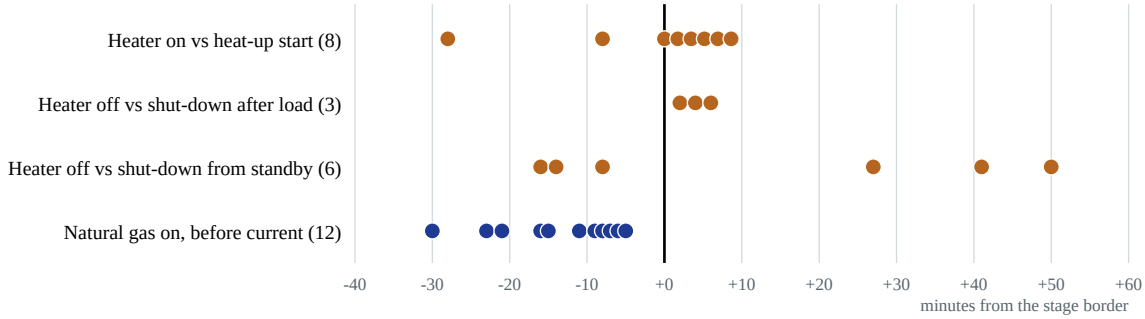
4 · What the current and power sensors add

They can't redefine the stages on their own. Both are already part of today's rules, and the current is zero for 99.5 % of the time. What they do well is **pin a border to the minute**: the heater marks where heating starts, the current marks where load starts and stops.



Schematic cycle with the heater and the current added. Dashed lines: the two forming-gas borders.

Border	Sensor that switches	How sharp, in the 39 days
1 Heat-up starts	Heater on	On the minute in 6 of 8 starts; 8 and 28 min early in two
2 Forming gas on	Gas	193-274 °C in 10 of 10 heat-ups
3 Ready for load	Natural gas on	Always 5-30 min before current (12 of 12)
4 Load starts	Current >= 1 A	Exact; today's Part load rule
5 Load ends	Current back to 0	Exact
6 Shut-down starts	Heater off	2-6 min after a load; from standby 16 min before to 50 min after
7 Forming gas off	Gas	210-292 °C in 5 of 6 cool-downs



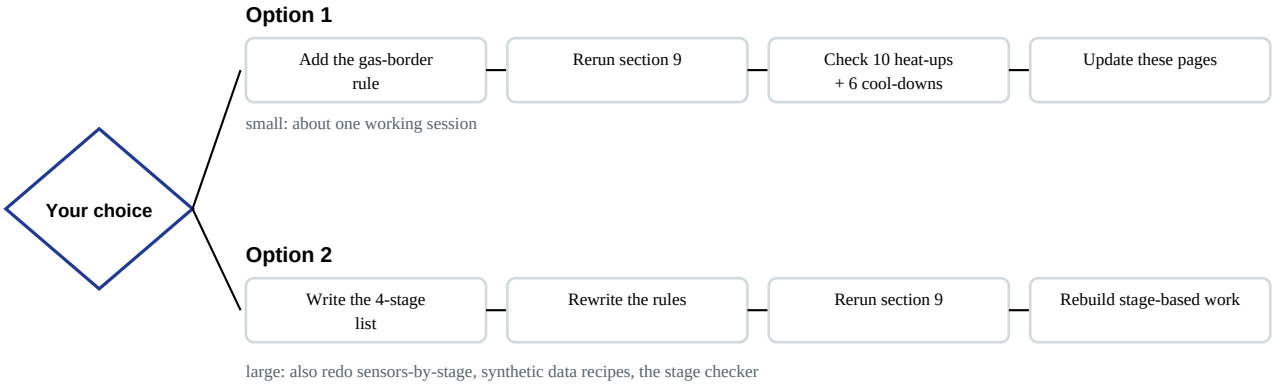
One dot per event; dots on the black line agree with the stage border to the minute. Shutting down from standby is the loose case: there the falling temperature, not the heater, has to decide.

String current (EIRC 41-45): already decides Part load (>= 1 A for 3 min); only 4.5 h of 913 h, peak 16.2 A per string; Rated power (28.5 A) never reached. **Stack voltage and power:** the voltage reads 0 except under load (~455 V), so it repeats the current; stack power peaks at 33.6 kW. **Heater power (EAPH):** on in nearly every hot minute; heating runs 15-20 kW, holding 9-14 kW, cool-back ~5 kW, but the ranges overlap, so the level can only support the temperature rule, not replace it.

If you remember one thing: current marks load and nothing else; the heater marks where heating starts. Neither can carry the stages alone, but together with the gas, every border can take the sensor that switches there.

5 · What happens after you choose

Option 1 is **four short steps**. Option 2 is also four steps, but the last one reopens every piece of work built on the stages.



"Check" means looking at every gas switch point in the logs: 10 heat-ups and 6 cool-downs. Each new border should land within a few minutes of the valve change.

Two small questions either way

The long forming-gas cool-down: on 2026-03-25 to 26 forming gas stayed on down to 69 °C. Suggestion: count the whole cool-down as under forming gas, following the valve as it was. **The short gas overlaps:** the 10-140 second overlaps stay a detail inside their stage; they are shorter than the 3-minute minimum.

Where the work happens: stage rules in data_analysis/operating_states.py (gas settings from line 60); figures in operating_states.ipynb, section 9.4b.

If you remember one thing: pick option 1 or option 2. The change itself is quick; the difference is how much of the later work has to be redone.